

- **7.7.15** When specified by the purchaser, external girth flanges shall be provided with 3 mm ($1/8$ in.) future machining allowance on the gasket contact seating surface (including pass-partition surfaces). The additional thickness shall not be used in the calculation of maximum allowable working pressure.

NOTE This requirement does not apply to clad or weld overlay construction.

7.7.16 Tongue and groove type configuration shall not be used, unless specified or agreed by the purchaser.

7.8 Girth Flange Joint Supplementary Design Requirements

- **7.8.1** The joint component approach, as defined in ASME PCC-1, Appendix O, and the requirements contained within this section, shall be applied to one or both sides of the heat exchanger, when specified by the purchaser. Definition of terms and symbology contained within this section are consistent with ASME PCC-1. See A.4.5 for additional guidance.

7.8.2 The minimum permissible bolt stress, $S_{b_{min}}$, and the maximum permissible bolt stress, $S_{b_{max}}$, shall be 40 % and 70 %, respectively, of the listed yield values in the pressure design code at assembly conditions (i.e. ambient temperature).

7.8.3 At least one-half of the gasket pass-partition rib area shall be added to the peripheral ring gasket area, A_g , unless otherwise agreed with the purchaser.

7.8.4 The selected assembly bolt stress, $S_{b_{sel}}$, shall be established so as to ensure that the gasket stresses comply with Table 3. For gasket types not listed, the minimum and maximum gasket stresses shall be agreed between the purchaser and vendor.

7.8.5 The fraction of gasket load remaining after relaxation, ϕ_g , used in the calculations shall be 0.7, unless otherwise specified by the purchaser.

7.8.6 To minimize potential for flange rotation, the rigidity index as calculated in ASME BPVC, Section VIII Division 1, Mandatory Appendix 2, shall not exceed 1.0.

7.8.7 For stationary tubesheets with gaskets on both the shell and tube sides of the tubesheet, in Step 6 of ASME PCC-1, the following additional checks, as calculated by Equation (1) and Equation (2), shall be undertaken to verify that the selected assembly bolt stress include the effects of pressure acting from the opposing side of the tubesheet. In this case, the maximum design pressure, P_{max} , shall be taken as the design pressure difference between the two sides.

Table 3—Assembly Gasket Stress

Peripheral Gasket Type ^a	Maximum Permissible Gasket Stress, $S_{g_{max}}$ MPa (psi)	Minimum Gasket Seating Stress, $S_{g_{min-S}}$ MPa (psi)	Minimum Gasket Operating Stress, $S_{g_{min-O}}$ MPa (psi)
Grooved metal with soft facing	380 (55,000)	140 (20,000)	97 (14,000)
Corrugated metal with soft facing	275 (40,000)	140 (20,000)	97 (14,000)
Spiral-wound	b	140 (20,000)	97 (14,000)

^a These stresses are based on the use of facing layers or filler materials, such as flexible graphite, PTFE, or other conformable materials, and excludes core materials of carbon steel, brass, copper, or aluminum alloys.

^b The maximum permissible gasket stress does not apply when a means to prevent over-compression of the gasket is employed (e.g. centering rings with spiral-wound gaskets).

Step 6a: Check selected assembly bolt stress is adequate for the shell side gasket:

$$Sb_{sel} \geq \frac{(Sg_{min-o} \times A_{g,S}) + \left[\frac{\pi}{4} \times (G_{ID,S})^2 \times (P_S - P_T) \right]}{(\phi_g \times n_b \times A_b)} \quad (1)$$

Step 6b: Check selected assembly bolt stress is adequate for the tube side gasket:

$$Sb_{sel} \geq \frac{(Sg_{min-o} \times A_{g,T}) + \left[\frac{\pi}{4} \times (G_{ID,T})^2 \times (P_T - P_S) \right]}{(\phi_g \times n_b \times A_b)} \quad (2)$$

where

- $A_{g,S}$ is the shell side gasket area, expressed in mm² (in.²);
- $A_{g,T}$ is the tube side gasket area, expressed in mm² (in.²);
- $G_{ID,S}$ is the shell side gasket inner diameter, expressed in mm (in.);
- $G_{ID,T}$ is the tube side gasket inner diameter, expressed in mm (in.);
- n_b is the number of bolts;
- P_S is the design pressure on the shell side, expressed in MPa (psi);
- P_T is the design pressure on the tube side, expressed in MPa (psi);
- Sb_{sel} is the selected assembly bolt stress, expressed in MPa (psi);
- Sg_{min-o} is the minimum gasket operating stress, expressed in MPa (psi);
- ϕ_g is the fraction of gasket load remaining after relaxation.

7.8.8 If the requirements contained within this section require the bolt area to be increased or gasket geometry revised, then the pressure design code calculations shall be updated and the required flange thickness re-calculated accordingly.

7.8.9 Controlled bolt-tightening procedures shall be applied, and records provided, to demonstrate that the selected assembly bolt stress, Sb_{sel} , has been achieved.

7.9 Expansion Joints

7.9.1 Expansion joints in shells shall be thin-wall bellows or thick walled (e.g. flanged, or flanged and flued). Floating head tail pipe expansion joints shall be thin wall bellows type.

7.9.2 Expansion joints made of thin-walled bellows shall:

- a) be protected from internal damage with a metallic liner mounted flush with the inside of the shell. The liner material shall be compatible with the base material to which it is attached. Carbon steel liners shall be at least